ines certify - a certification process for energy system planning models using the ines-spec

Documentation written for the European project Mopo

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Chapter 1

Proof of concept for the certification process

The ines-spec describes an interoperable data format that is to be used to ease the communication between different energy system planning models and corresponding input data sets. Instead of multiple conversion scripts between multiple tools, every tool only needs a conversion to and from the ines-spec. To verify the correct working of these scripts, we set up a certification process. At the same time, the certification process allows to verify the proper implementation of certain features in energy system planning models such as, e.g., reserves.

The certification process consists of a few predefined setups of which the exact solution is known as the optimal solution can be calculated manually. In practice, instead of providing a manual derivation here, we have a custom model for each setup implemented in Julia JuMP. The solutions are still calculated manually to check these custom models but these manual calculations are not documented. The same setup is then also implemented in the ines-spec as well as the original format of the energy system planning model. The conversion script needs to be capable of translating the setup in the ines-spec to the same setup in the energy system planning model. The result of the energy system planning model then also needs to correspond to the solution of the exact solution.

The current state of the certification process is a proof of concept with a select number of setups applied to a single model, i.e. SpineOpt. Of course, SpineOpt succeeds for all of these setups as they have been iteratively aligned during the creation of the proof of concept. The goal is to show the potential of the certification process with the ines-spec. Once a consortium has been formed around the ines-spec, the certification process can be further developed and improved.

In the meantime, the certification process is ready to be adopted by the community. The setups with the ines spec as well as the exact solution remain the same for every tool. Each tool only needs to recreate the setups in their respective format. These setups can then be used to certify the tool directly by comparing the results of the tool with the exact solution or to create/verify the conversion tool from/to ines.

In the following, we describe the repository organisation and the requirements for the workflow in Spine Toolbox. Thereafter, we provide a small overview of all the setups and go into more detail in each of these. In the appendix you can find a non exhaustive overview of additional setups that can be implemented in the future.

Repository organisation and requirements

There is a separate repository for the certification process. However, this repository is only supposed to hold tools and scripts for automation and benchmarking. Automation and benchmarking are not part of the scope of the proof of concept, so currently the repository is just a placeholder. The actually important files are in the ines-spec repository and the repositories of the conversion scripts. The files are always in an 'ines certify' folder inside these repositories. The ines-spec repository holds the setup in the ines format as well as the corresponding julia script to compute the exact solution. The repositories for the conversion scripts hold the equivalent setups in the format of the energy system planning model, e.g. SpineOpt.

To get the certification process running with SpineOpt, you need to install Spine Toolbox and clone the SpineOpt, SpineInterface, ines-tools, ines-spec, ines-spineopt repositories. More details on the installation and use can be found in the appendix B.

For a new model to join the certification process, they need to create the setups in their own model. The conversion script between the ines-spec and the tool can then be calibrated based on these setups. The results of the model can then be compared to the exact solution for the certification.

Overview

Figure 1.1 shows the workflow in Spine Toolbox and Table 1.1 gives a brief overview of the setups. Essentially the workflow consists of a spine database for every setup of the certification process in the ines-spec. These are fed to the ines to, e.g., SpineOpt conversion script. After that there is a database for the SpineOpt input data that is preloaded with the SpineOpt template (before the conversion script is used). That input is fed to the energy system planning model, e.g. SpineOpt and the results are stored in another spine database. These results can then be compared manually to the results of the exact solution that is calculated in Julia JuMP outside of Spine Toolbox.

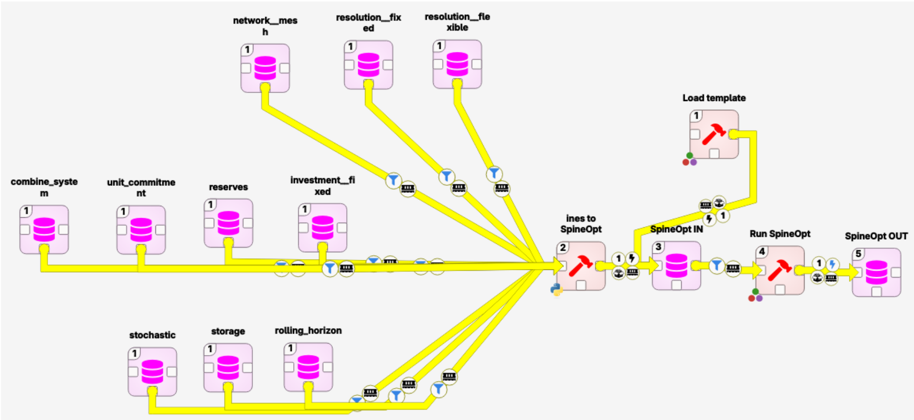

Figure 1.1: Typical workflow for the certification process.

There are 10 setups in Table 1.1, 9 of which are considered to be unit tests and 1 is a system test that combines some of the unit tests. Each unit tests focuses on a single feature of energy system planning models. Some features are more complex and therefore build on top of a smaller unit test. The tests are cherry picked to cover some of the more prevalent aspects of energy system planning models, including model detail (unit commitment), operation and investments as well as the stochastic and temporal structure. Algorithms are more difficult to deal with as the solutions can more easily deviate from the exact solution. It is possible to introduce a margin for those setups but that is outside the scope of the proof of concept. We do include an algorithm that is typically more exact, i.e. a simple rolling horizon. Note that all of the setups are rather artificial. The simple reason is that strong simplifications are necessary to make the manual calculations more easy. For example, an efficiency of 0.8 is chosen over an efficiency of 0.9 as the fractions with 0.8 are much easier to deal with. In the following we go more into detail of all of these setups. For each setup there is a small description, a picture of the setup in the ines-spec and the corresponding formulation for the custom model with the exact solution. A legend of the symbols used in the pictures can be found in appendix A. The symbols in the formulation are supposed to be self explanatory as they correspond to the parameters fo the ines-spec (in addition, the letter p stands for parameter and the letter v stands for variable).

Table 1.1: Overview of the setups for the certification process.

name	description
network = meshed	Energy balance where the demand equals the supply for a single time step. Triangular (but one directional) network with the two supply units and demand unit. The cheapest unit should be chosen.
resolution = fixed	network = meshed but expanded to a time series. Both units have the same resolution.
resolution = flexible	resolution = fixed but the two units have different resolutions (1h for expensive unit and 2h for cheap unit).
stochastic	resolution = fixed but there are scenarios for the demand and the resource prices.
reserves	resolution = fixed but there is a demand for reserves. The expensive unit is able to provide reserves but not enough to entirely avoid load shedding.
unit commitment	resolution = fixed but there is unit commitment for both units. There is a minimum uptime, a minimum downtime and a part-load efficiency.
investment = fixed	resolution = fixed but there are investment variables.
storage	resolution = fixed but with storage and renewable unit.
rolling horizon	investment = fixed but there are two horizons and the model rolls from one horizon to the other. The demand is increased in the second horizon.
combined system	A combination of most the previous setups. Sector coupled for the flexible time steps. The gas sector consists of import, demand and gas provision to a gas plant. The electricity sector consists of the gas plant, a nuclear plant, batteries, onshore PV, offshore wind, electricity import and demand. The demand is in the distribution grid, the import and offshore wind is in a separate node. There are stochastics for the demand.

1.1 network = mesh

The goal of the setup is to test the energy balance in space. To that end the demand and supply are scattered over different nodes. They are connected to each other through directional lines with small losses but sufficient capacity. The lines are directional to avoid circular flows and the lines have losses to avoid multiple possible solutions. One of the supply units is obviously cheaper such that it is very clear what the solution should be.

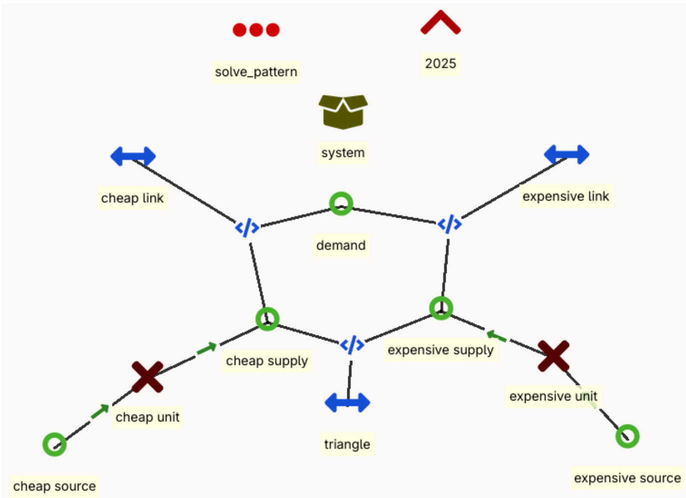

Figure 1.2: Implementation of network = mesh in the ines-spec.

minimise:

$$z = p^{other\_operational\_costs}_{cheap\_source} \cdot v^{flow}_{cheap\_source} + p^{other\_operational\_costs}_{expensive\_source} \cdot v^{flow}_{expensive\_source}$$

subject to:

$$\begin{aligned}
p^{flow_profile}_{demand} &= v^{flow}_{cheap_link,out} + v^{flow}_{expensive_link,out} \\
v^{flow}_{cheap_link,out} &= p^{efficiency}_{cheap_link} \cdot v^{flow}_{cheap_link,in} \\
v^{flow}_{expensive_link,out} &= p^{efficiency}_{expensive_link} \cdot v^{flow}_{expensive_link,in} \\
v^{flow}_{triangle,out} &= p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in} \\
v^{flow}_{cheap_link,in} &= v^{flow}_{cheap_supply} + v^{flow}_{triangle,out} \\
v^{flow}_{expensive_supply} &= v^{flow}_{expensive_link,in} + v^{flow}_{triangle,in} \\
v^{flow}_{expensive_supply} &= p^{efficiency}_{expensive_unit} \cdot v^{flow}_{expensive_source} \\
v^{flow}_{cheap_supply} &= p^{efficiency}_{cheap_unit} \cdot v^{flow}_{cheap_source}
\end{aligned}$$

bound to:

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_link,in} \\
0 \leq & v^{flow}_{cheap\_link,out} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,in} \\
0 \leq & v^{flow}_{expensive\_link,out} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,in} \\
0 \leq & v^{flow}_{triangle,out} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
0 \leq & v^{flow}_{expensive\_supply} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \leq & v^{flow}_{expensive\_source} \\
0 \leq & v^{flow}_{cheap\_supply} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
0 \leq & v^{flow}_{cheap\_source}
\end{aligned}
$$

1.2 resolution = fixed

The goal of the setup is to test the energy balance in time. To that end, the setup starts from network = meshed and adds a time series for the demand and the supply. All entities have the same resolution.

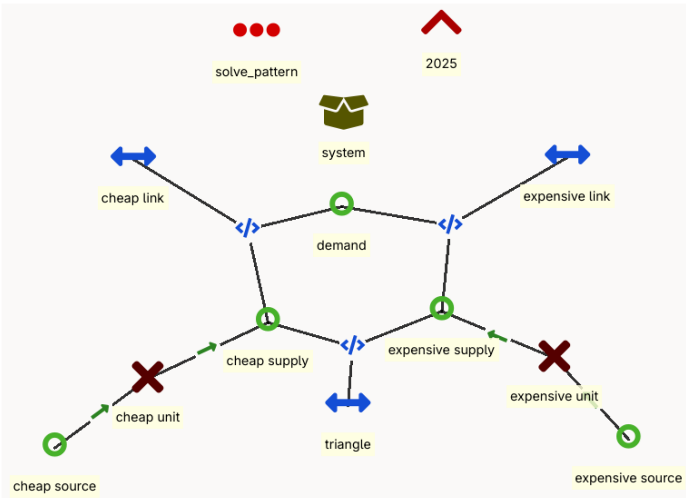

Figure 1.3: Implementation of resolution = fixed in the ines-spec.

minimise:

$$
z = \sum_t \{ p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t} \}
$$

subject to:

$\forall t$

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} =& v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t}\\
v^{flow}_{cheap\_link,out,t} =& p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t}\\
v^{flow}_{expensive\_link,out,t} =& p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t}\\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t}\\
v^{flow}_{cheap\_link,in,t} =& v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t}\\
v^{flow}_{expensive\_supply,t} =& v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t}\\
v^{flow}_{expensive\_supply,t} =& p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t}\\
v^{flow}_{cheap\_supply,t} =& p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t}
\end{aligned}
$$

bound to:

$\forall t$

$$
\begin{aligned}
0 \leq& v^{flow}_{cheap\_link,in,t}\\
0 \leq& v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link}\\
0 \leq& v^{flow}_{expensive\_link,in,t}\\
0 \leq& v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link}\\
0 \leq& v^{flow}_{triangle,in,t}\\
0 \leq& v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle}\\
0 \leq& v^{flow}_{expensive\_supply,t} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit}\\
0 \leq& v^{flow}_{expensive\_source,t}\\
0 \leq& v^{flow}_{cheap\_supply,t} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit}\\
0 \leq& v^{flow}_{cheap\_source,t}
\end{aligned}
$$

1.3 resolution = flexible

The goal of this setup is to test the energy balance in time and space. To that end, the setup starts from resolution = fixed but reduces the resolution of one of the supply units. The resolution is only changed at the source such that the supply of the unit is still able to provide the same output as before.

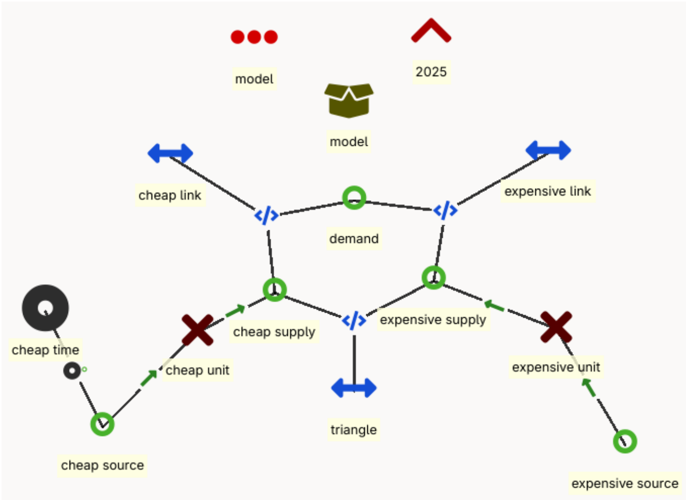

Figure 1.4: Implementation of resolution = flexible in the ines-spec. The setup is similar as befor but there is a time set with a different resolution attached to the cheap source.

minimise:

$$z = \sum_{t \in t_{cheap}} \{p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} \cdot \Delta_{t_{cheap}}\} + \sum_{t} \{p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t}\}$$

subject to:

$\forall t$

$$\begin{aligned}
p^{flow_profile}_{demand,t} =& v^{flow}_{cheap_link,out,t} + v^{flow}_{expensive_link,out,t} \\
v^{flow}_{cheap_link,out,t} =& p^{efficiency}_{cheap_link} \cdot v^{flow}_{cheap_link,in,t} \\
v^{flow}_{expensive_link,out,t} =& p^{efficiency}_{expensive_link} \cdot v^{flow}_{expensive_link,in,t} \\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t} \\
v^{flow}_{cheap_link,in,t} =& v^{flow}_{cheap_supply,t} + v^{flow}_{triangle,out,t} \\
v^{flow}_{expensive_supply,t} =& v^{flow}_{expensive_link,in,t} + v^{flow}_{triangle,in,t} \\
v^{flow}_{expensive_supply,t} =& p^{efficiency}_{expensive_unit} \cdot v^{flow}_{expensive_source,t}
\end{aligned}$$

$\forall t_{cheap}$

$$\sum_{t \in [t_{cheap}:t_{cheap}+\Delta_{t_{cheap}}]} v^{flow}_{cheap\_supply,t} \cdot 1h = p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t_{cheap}} \cdot \Delta_{t_{cheap}}$$

bound to:

$\forall t$

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_link,in,t} \\
0 \leq & v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,in,t} \\
0 \leq & v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,in,t} \\
0 \leq & v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
0 \leq & v^{flow}_{expensive\_supply,t} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \leq & v^{flow}_{expensive\_source,t} \\
0 \leq & v^{flow}_{cheap\_supply,t} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit}
\end{aligned}
$$

$\forall t_{cheap}$

$$
0 \leq v^{flow}_{cheap\_source,t_{cheap}} \qquad \leq \frac{1}{p^{efficiency}_{cheap\_unit}} \cdot p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit}
$$

1.4 stochastic

The goal of this setup is to test scenarios in the formulation. To that end the setup starts from resolution = fixed and adds stochastics for the demand and the commodity prices. The demand can be high and low and the commodity prices can be expensive or cheap. Combining these scenarios, results in 4 equally likely scenarios for the entire model.

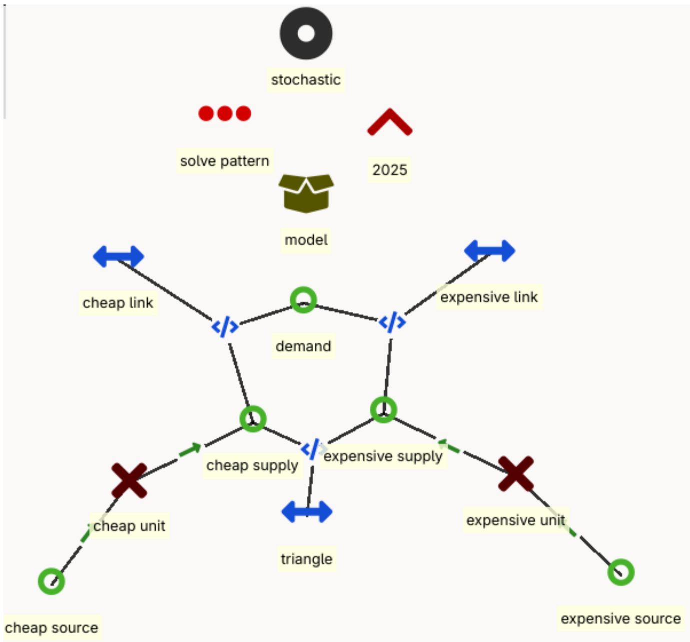

Figure 1.5: Implementation of stochastic in the ines-spec.

minimise:

$$z = \sum_{s} \{p_{s}^{weight} \cdot \sum_{t} \{p_{cheap\_source,t,s}^{other\_operational\_costs} \cdot v_{cheap\_source,t,s}^{flow} + p_{expensive\_source,t,s}^{other\_operational\_costs} \cdot v_{expensive\_source,t,s}^{flow}\}\}$$

subject to:

$\forall t, \forall s$

$$\begin{aligned}
v_{cheap_link,out,t,s}^{flow} =& p_{cheap_link}^{efficiency} \cdot v_{cheap_link,in,t,s}^{flow} \\
v_{expensive_link,out,t,s}^{flow} =& p_{expensive_link}^{efficiency} \cdot v_{expensive_link,in,t,s}^{flow} \\
v_{triangle,out,t,s}^{flow} =& p_{triangle}^{efficiency} \cdot v_{triangle,in,t,s}^{flow} \\
v_{cheap_link,in,t,s}^{flow} =& v_{cheap_supply,t,s}^{flow} + v_{triangle,out,t,s}^{flow} \\
v_{expensive_supply,t,s}^{flow} =& v_{expensive_link,in,t,s}^{flow} + v_{triangle,in,t,s}^{flow} \\
v_{expensive_supply,t,s}^{flow} =& p_{expensive_unit}^{efficiency} \cdot v_{expensive_source,t,s}^{flow} \\
v_{cheap_supply,t,s}^{flow} =& p_{cheap_unit}^{efficiency} \cdot v_{cheap_source,t,s}^{flow}
\end{aligned}$$

bound to:

$\forall t, \forall s$

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_link,in,t,s} \\
0 \leq & v^{flow}_{cheap\_link,out,t,s} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,in,t,s} \\
0 \leq & v^{flow}_{expensive\_link,out,t,s} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,in,t,s} \\
0 \leq & v^{flow}_{triangle,out,t,s} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
0 \leq & v^{flow}_{expensive\_supply,t,s} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \leq & v^{flow}_{expensive\_source,t,s} \\
0 \leq & v^{flow}_{cheap\_supply,t,s} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
0 \leq & v^{flow}_{cheap\_source,t,s}
\end{aligned}
$$

1.5 reserves

The goal of this setup is to test the reserve requirements. To that end the setup starts from resolution = fixed and adds a reserve requirement to the demand. The reserves can be provided by the two units but there is also the possibility of load shedding (through a slack variable of the demand). A common way to include reserves is by working with a reserve node, dismissing the capacity restrictions of the lines. For the proof of concept, we limit ourselves to upward reserves.

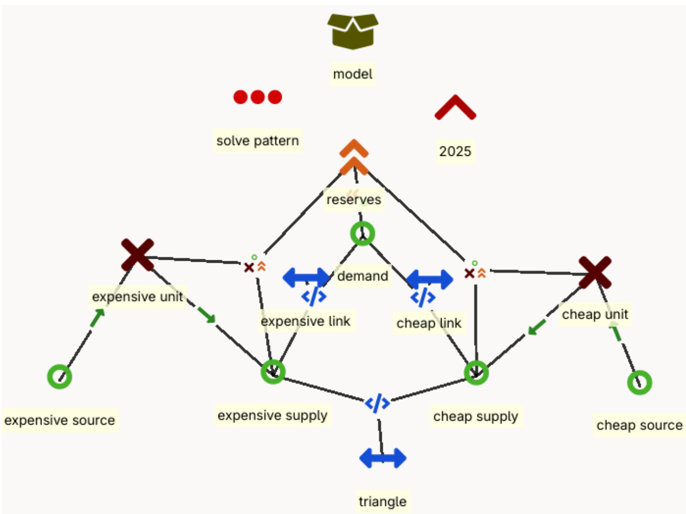

Figure 1.6: Implementation of reserves in the ines-spec. There is a separate reserve entity which is connected to the cheap unit, the expensive unit and the demand.

minimise:

$$
\begin{aligned}
z = &\sum_t \{ p_{cheap\_source,t}^{other\_operational\_costs} \cdot v_{cheap\_source,t}^{flow} + p_{expensive\_source,t}^{other\_operational\_costs} \cdot v_{expensive\_source,t}^{flow} \} \\
&+ \sum_t \{ p_{demand}^{penalty} \cdot v_{demand,t}^{slack} + p_{cheap\_supply,t}^{procurement\_cost} \cdot v_{cheap\_supply,t}^{flow,reserve} + p_{expensive\_supply,t}^{procurement\_cost} \cdot v_{expensive\_supply,t}^{flow,reserve} \}
\end{aligned}
$$

subject to:

$\forall t$

$$
p_{demand,t}^{reserve} =< v_{cheap\_supply,t}^{flow,reserve} + v_{expensive\_supply,t}^{flow,reserve} + v_{demand,t}^{slack}
$$

$$
p_{demand,t}^{flow\_profile} = v_{cheap\_link,out,t}^{flow} + v_{expensive\_link,out,t}^{flow}
$$

$$
\begin{aligned}
v_{cheap\_link,out,t}^{flow} &= p_{cheap\_link}^{efficiency} \cdot v_{cheap\_link,in,t}^{flow} \\
v_{expensive\_link,out,t}^{flow} &= p_{expensive\_link}^{efficiency} \cdot v_{expensive\_link,in,t}^{flow} \\
v_{triangle,out,t}^{flow} &= p_{triangle}^{efficiency} \cdot v_{triangle,in,t}^{flow}
\end{aligned}
$$

$$
\begin{aligned}
v_{cheap\_link,in,t}^{flow} &= v_{cheap\_supply,t}^{flow} + v_{triangle,out,t}^{flow} \\
v_{expensive\_supply,t}^{flow} &= v_{expensive\_link,in,t}^{flow} + v_{triangle,in,t}^{flow}
\end{aligned}
$$

$$
\begin{aligned}
v_{expensive\_supply,t}^{flow} &= p_{expensive\_unit}^{efficiency} \cdot v_{expensive\_source,t}^{flow} \\
v_{cheap\_supply,t}^{flow} &= p_{cheap\_unit}^{efficiency} \cdot v_{cheap\_source,t}^{flow}
\end{aligned}
$$

$$
\begin{aligned}
v_{expensive\_supply,t}^{flow} + v_{expensive\_supply,t}^{flow,reserve} &\leq p_{expensive\_unit}^{capacity} \cdot p_{expensive\_unit}^{existing\_units} \\
v_{cheap\_supply,t}^{flow} + v_{expensive\_supply,t}^{flow,reserve} &\leq p_{cheap\_unit}^{capacity} \cdot p_{cheap\_unit}^{existing\_units}
\end{aligned}
$$

bound to:

$\forall t$

$$
\begin{aligned}
0 \leq & v^{slack}_{demand,t} \\
0 \leq & v^{flow,reserve}_{cheap\_supply,t} && \leq p^{max\_reserve}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
0 \leq & v^{flow,reserve}_{expensive\_supply,t} && \leq p^{max\_reserve}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
\\
0 \leq & v^{flow}_{cheap\_link,in,t} \\
0 \leq & v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,in,t} \\
0 \leq & v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,in,t} \\
0 \leq & v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
\\
0 \leq & v^{flow}_{expensive\_supply,t} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \\
0 \leq & v^{flow}_{expensive\_source,t} \\
0 \leq & v^{flow}_{cheap\_supply,t} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \\
0 \leq & v^{flow}_{cheap\_source,t}
\end{aligned}
$$

1.6 unit commitment

The goal of this setup is to test the unit commitment functionality. To that end, the setup starts from resolution = fixed but adds on/off state variables to the supply units. The commitment is constrained by the minimum up and downtime and there is a cost associated to starting units. There are no initial conditions for these constraints as the model can choose the ideal starting point for the units that are on. The underlying assumption is that the system was already running before the simulation and that the demand was similar to the demand in the first step.

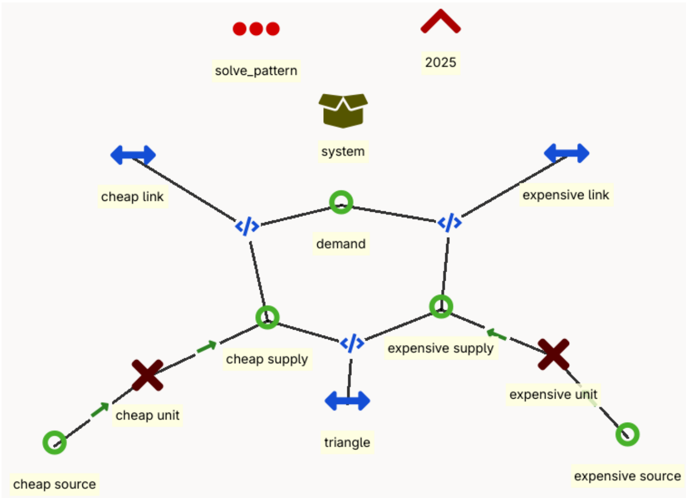

Figure 1.7: Implementation of unit commitment in the ines-spec.

minimise:

$$
\begin{aligned}
z = \sum_t \{ & p^{startup\_cost}_{cheap\_unit} \cdot v^{on\_up.}_{cheap\_unit,t} + p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} \\
& + p^{startup\_cost}_{expensive\_unit} \cdot v^{on\_up.}_{expensive\_unit,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t} \}
\end{aligned}
$$

subject to:

$\forall t$

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} =& v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,out,t} =& p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t}\\
v^{flow}_{expensive\_link,out,t} =& p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t}\\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,in,t} =& v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t}\\
v^{flow}_{expensive\_supply,t} =& v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{expensive\_supply,t} =& p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t}\\
v^{flow}_{cheap\_supply,t} =& p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{on}_{cheap\_unit,t} =& v^{on}_{cheap\_unit,t-1} + v^{on\_up}_{cheap\_unit,t} - v^{on\_down}_{cheap\_unit,t}\\
v^{on}_{expensive\_unit,t} =& v^{on}_{expensive\_unit,t-1} + v^{on\_up}_{expensive\_unit,t} - v^{on\_down}_{expensive\_unit,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_supply,t} \leq& p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit} \cdot v^{on}_{cheap\_unit,t}\\
v^{flow}_{expensive\_supply,t} \leq& p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit} \cdot v^{on}_{expensive\_unit,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{on\_up}_{cheap\_unit,t} \leq& 1 - v^{on}_{cheap\_unit,t} - \sum_{t'=1}^{MDT-1} v^{on\_down}_{cheap\_unit,t-t'}\\
v^{on\_down}_{cheap\_unit,t} \leq& v^{on}_{cheap\_unit,t} - \sum_{t'=1}^{MUT-1} v^{on\_up}_{cheap\_unit,t-t'}\\
v^{on\_up}_{expensive\_unit,t} \leq& 1 - v^{on}_{expensive\_unit,t} - \sum_{t'=1}^{MDT-1} v^{on\_down}_{expensive\_unit,t-t'}\\
v^{on\_down}_{expensive\_unit,t} \leq& v^{on}_{expensive\_unit,t} - \sum_{t'=1}^{MUT-1} v^{on\_up}_{expensive\_unit,t-t'}
\end{aligned}
$$

bound to:

$\forall t$

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_link,in,t} \\
0 \leq & v^{flow}_{cheap\_link,out,t} & \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,in,t} \\
0 \leq & v^{flow}_{expensive\_link,out,t} & \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,in,t} \\
0 \leq & v^{flow}_{triangle,out,t} & \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq & v^{flow}_{expensive\_supply,t} \\
0 \leq & v^{flow}_{expensive\_source,t}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_supply,t} \\
0 \leq & v^{flow}_{cheap\_source,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{on}_{cheap\_unit,t} &\in \{0,1\} \\
v^{on\_up}_{cheap\_unit,t} &\in \{0,1\} \\
v^{on\_down}_{cheap\_unit,t} &\in \{0,1\} \\
v^{on}_{expensive\_unit,t} &\in \{0,1\} \\
v^{on\_up}_{expensive\_unit,t} &\in \{0,1\} \\
v^{on\_down}_{expensive\_unit,t} &\in \{0,1\}
\end{aligned}
$$

1.7 investments = fixed

The goal of this setup is to test basic investments. To that end the resolution = fixed setup is expanded with investment variables. The expensive unit is already invested in to incentivise the model to keep using the expensive unit. In the formulation below, for brevity, the investment cost is to be interpreted as the cost per unit, as opposed to the cost per MW(h).

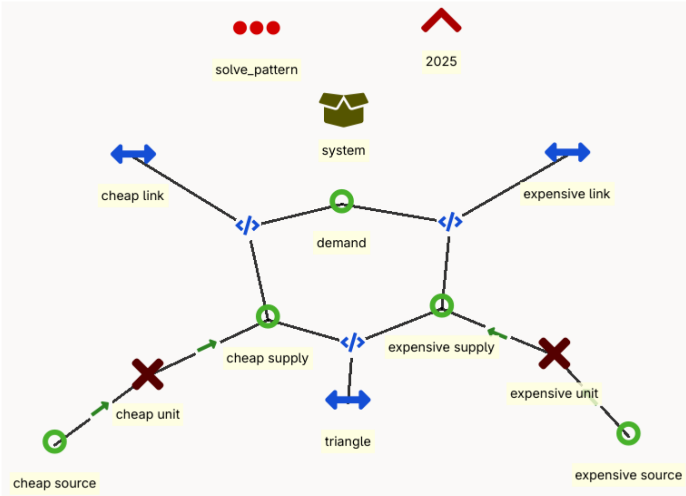

Figure 1.8: Implementation of investment = fixed in the ines-spec.

minimise:

$$\begin{aligned}
z =& p^{investment_cost}_{cheap_unit} \cdot v^{investment}_{cheap_unit} + p^{investment_cost}_{expensive_unit} \cdot (v^{investment}_{expensive_unit} - p^{existing_units}_{expensive_unit}) \\
&+ \sum_t \{p^{other_operational_costs}_{cheap_source,t} \cdot v^{flow}_{cheap_source,t} + p^{other_operational_costs}_{expensive_source,t} \cdot v^{flow}_{expensive_source,t}\}
\end{aligned}$$

subject to:

$\forall t$

$$p^{flow\_profile}_{demand,t} = v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t}$$

$$\begin{aligned}
v^{flow}_{cheap_link,out,t} =& p^{efficiency}_{cheap_link} \cdot v^{flow}_{cheap_link,in,t} \\
v^{flow}_{expensive_link,out,t} =& p^{efficiency}_{expensive_link} \cdot v^{flow}_{expensive_link,in,t} \\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t}
\end{aligned}$$

$$\begin{aligned}
v^{flow}_{cheap_link,in,t} =& v^{flow}_{cheap_supply,t} + v^{flow}_{triangle,out,t} \\
v^{flow}_{expensive_supply,t} =& v^{flow}_{expensive_link,in,t} + v^{flow}_{triangle,in,t}
\end{aligned}$$

$$\begin{aligned}
v^{flow}_{expensive_supply,t} =& p^{efficiency}_{expensive_unit} \cdot v^{flow}_{expensive_source,t} \\
v^{flow}_{cheap_supply,t} =& p^{efficiency}_{cheap_unit} \cdot v^{flow}_{cheap_source,t}
\end{aligned}$$

$$\begin{aligned}
v^{flow}_{expensive_supply,t} \le& p^{capacity}_{expensive_unit} \cdot v^{investment}_{expensive_unit} \\
v^{flow}_{cheap_supply,t} \le& p^{capacity}_{cheap_unit} \cdot v^{investment}_{cheap_unit}
\end{aligned}$$

bound to:

$$
\begin{aligned}
0 \leq & v^{investment}_{cheap\_unit} \\
p^{existing\_units}_{expensive\_unit} \leq & v^{investment}_{expensive\_unit}
\end{aligned}
$$

$\forall t$

$$
\begin{aligned}
0 \leq & v^{flow}_{cheap\_link,in,t} \\
0 \leq & v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 \leq & v^{flow}_{expensive\_link,in,t} \\
0 \leq & v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 \leq & v^{flow}_{triangle,in,t} \\
0 \leq & v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
0 \leq & v^{flow}_{expensive\_supply,t} \\
0 \leq & v^{flow}_{expensive\_source,t} \\
0 \leq & v^{flow}_{cheap\_supply,t} \\
0 \leq & v^{flow}_{cheap\_source,t}
\end{aligned}
$$

1.8 storage

The goal of this setup is to test an ideal storage formulation. To that end, storage and a renewable unit are added to the resolution = fixed setup. The storage unit is located at the cheap supply side and the renewable unit is located at the expensive supply side. Storage starts empty.

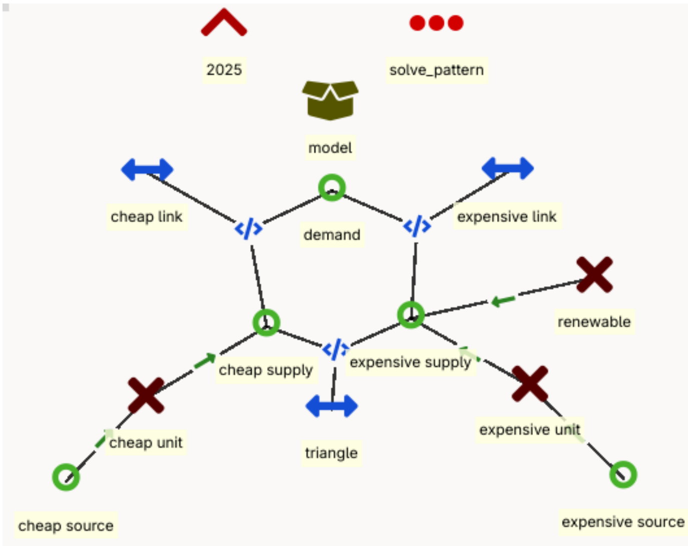

Figure 1.9: Implementation of storage in the ines-spec. Storage is located at the node of the cheap supply.

minimise:

$$z = \sum_t \{p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t}\}$$

subject to:

$$v^{state}_{storage,1} = p^{efficiency,charge}_{storage} \cdot v^{charge}_{storage,1} - v^{discharge}_{storage,1}$$

$\forall t$

$$
\begin{aligned}
v^{state}_{storage,t} =& p^{efficiency}_{storage} \cdot v^{state}_{storage,t-1} + p^{efficiency,charge}_{storage} \cdot v^{charge}_{storage,t} - v^{discharge}_{storage,t}\\
\\
p^{flow\_profile}_{demand,t} =& v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t}\\
\\
v^{flow}_{cheap\_link,out,t} =& p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t}\\
v^{flow}_{expensive\_link,out,t} =& p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t}\\
v^{flow}_{triangle,out,t} =& p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t}\\
\\
v^{flow}_{cheap\_link,in,t} + v^{charge}_{storage,t} =& v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t} + p^{efficiency,discharge}_{storage} \cdot v^{discharge}_{storage,t}\\
v^{flow}_{expensive\_supply,t} + v^{flow}_{renewable,t} =& v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t}\\
\\
v^{flow}_{expensive\_supply,t} =& p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t}\\
v^{flow}_{cheap\_supply,t} =& p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t}
\end{aligned}
$$

bound to:

$\forall t$

$$
\begin{aligned}
0 \leq& v^{flow}_{cheap\_link,in,t}\\
0 \leq& v^{flow}_{cheap\_link,out,t} && \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link}\\
0 \leq& v^{flow}_{expensive\_link,in,t}\\
0 \leq& v^{flow}_{expensive\_link,out,t} && \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link}\\
0 \leq& v^{flow}_{triangle,in,t}\\
0 \leq& v^{flow}_{triangle,out,t} && \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle}\\
\\
0 \leq& v^{flow}_{cheap\_source,t}\\
0 \leq& v^{flow}_{cheap\_supply,t} && \leq p^{capacity}_{cheap\_unit} \cdot p^{existing\_units}_{cheap\_unit}\\
0 \leq& v^{flow}_{expensive\_source,t}\\
0 \leq& v^{flow}_{expensive\_supply,t} && \leq p^{capacity}_{expensive\_unit} \cdot p^{existing\_units}_{expensive\_unit}\\
\\
0 \leq& v^{flow}_{renewable,t} && \leq p^{flow\_profile}_{renewable,t} \cdot p^{capacity}_{renewable} \cdot p^{existing\_units}_{renewable}\\
\\
0 \leq& v^{charge}_{storage,t} && \leq p^{capacity\_charge}_{storage} \cdot p^{existing\_units}_{storage}\\
0 \leq& v^{discharge}_{storage,t} && \leq p^{capacity\_discharge}_{storage} \cdot p^{existing\_units}_{storage}\\
0 \leq& v^{state}_{storage,t} && \leq p^{capacity}_{storage} \cdot p^{existing\_units}_{storage}
\end{aligned}
$$

1.9 rolling horizon

The goal of the setup is to test a rolling horizon. To that end the investment = fixed setup is expanded with a second horizon. Each horizon uses the same formulation. In the first horizon, there is already a small investment in expensive units and the model needs to invest in cheaper units. The second window has a higher demand requiring additional investments but there is also a shift in commodity prices such that the more 'expensive' unit becomes more attractive, yet there are already the investments from the first window in the 'cheaper' units.

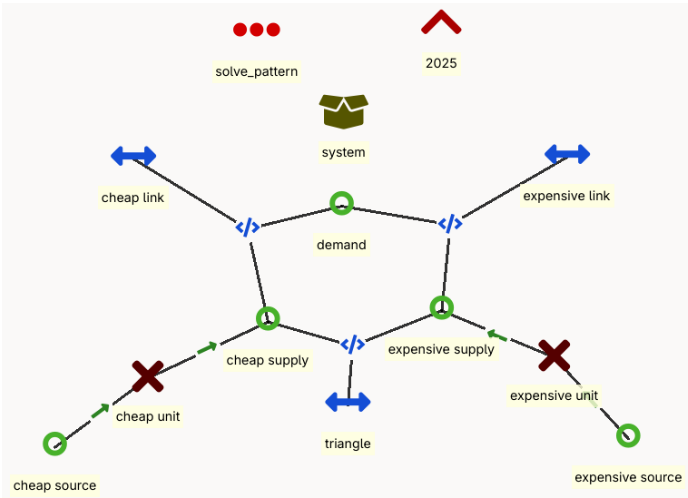

Figure 1.10: Implementation of unit commitment in the ines-spec.

minimise:

$$
\begin{aligned}
z = & p^{investment\_cost}_{cheap\_unit} \cdot (v^{investment}_{cheap\_unit} - p^{existing\_units}_{cheap\_unit}) + p^{investment\_cost}_{expensive\_unit} \cdot (v^{investment}_{expensive\_unit} - p^{existing\_units}_{expensive\_unit}) \\
& + \sum_t \{ p^{other\_operational\_costs}_{cheap\_source,t} \cdot v^{flow}_{cheap\_source,t} + p^{other\_operational\_costs}_{expensive\_source,t} \cdot v^{flow}_{expensive\_source,t} \}
\end{aligned}
$$

subject to:

$\forall t$

$$
\begin{aligned}
p^{flow\_profile}_{demand,t} &= v^{flow}_{cheap\_link,out,t} + v^{flow}_{expensive\_link,out,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,out,t} &= p^{efficiency}_{cheap\_link} \cdot v^{flow}_{cheap\_link,in,t} \\
v^{flow}_{expensive\_link,out,t} &= p^{efficiency}_{expensive\_link} \cdot v^{flow}_{expensive\_link,in,t} \\
v^{flow}_{triangle,out,t} &= p^{efficiency}_{triangle} \cdot v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{cheap\_link,in,t} &= v^{flow}_{cheap\_supply,t} + v^{flow}_{triangle,out,t} \\
v^{flow}_{expensive\_supply,t} &= v^{flow}_{expensive\_link,in,t} + v^{flow}_{triangle,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{expensive\_supply,t} &= p^{efficiency}_{expensive\_unit} \cdot v^{flow}_{expensive\_source,t} \\
v^{flow}_{cheap\_supply,t} &= p^{efficiency}_{cheap\_unit} \cdot v^{flow}_{cheap\_source,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{expensive\_supply,t} &\leq p^{capacity}_{expensive\_unit} \cdot v^{investment}_{expensive\_unit} \\
v^{flow}_{cheap\_supply,t} &\leq p^{capacity}_{cheap\_unit} \cdot v^{investment}_{cheap\_unit}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
p^{existing\_units}_{cheap\_unit} &\leq v^{investment}_{cheap\_unit} \\
p^{existing\_units}_{expensive\_unit} &\leq v^{investment}_{expensive\_unit}
\end{aligned}
$$

$\forall t$

$$
\begin{aligned}
0 &\leq v^{flow}_{cheap\_link,in,t} \\
0 &\leq v^{flow}_{cheap\_link,out,t} \leq p^{capacity}_{cheap\_link} \cdot p^{existing\_units}_{cheap\_link} \\
0 &\leq v^{flow}_{expensive\_link,in,t} \\
0 &\leq v^{flow}_{expensive\_link,out,t} \leq p^{capacity}_{expensive\_link} \cdot p^{existing\_units}_{expensive\_link} \\
0 &\leq v^{flow}_{triangle,in,t} \\
0 &\leq v^{flow}_{triangle,out,t} \leq p^{capacity}_{triangle} \cdot p^{existing\_units}_{triangle} \\
0 &\leq v^{flow}_{expensive\_supply,t} \\
0 &\leq v^{flow}_{expensive\_source,t} \\
0 &\leq v^{flow}_{cheap\_supply,t} \\
0 &\leq v^{flow}_{cheap\_source,t}
\end{aligned}
$$

1.10 combined system

The goal of this setup is to test systems that combine some of the different features. This setup is a combination of all the previous setups (except for the stochastics and rolling horizon to reduce the complexity for calculations by hand). It is sector coupled for the flexible time steps. The gas sector consists of import, demand and gas provision to a gas plant. The electricity sector consists of the gas plant, a nuclear plant, batteries, onshore PV, offshore wind, electricity import and demand. The demand is in the distribution grid, the import and offshore wind is in a separate node. The nuclear plant, batteries and import are already there, but the model can invest in a limited amount of gas plants and renewable energy systems.

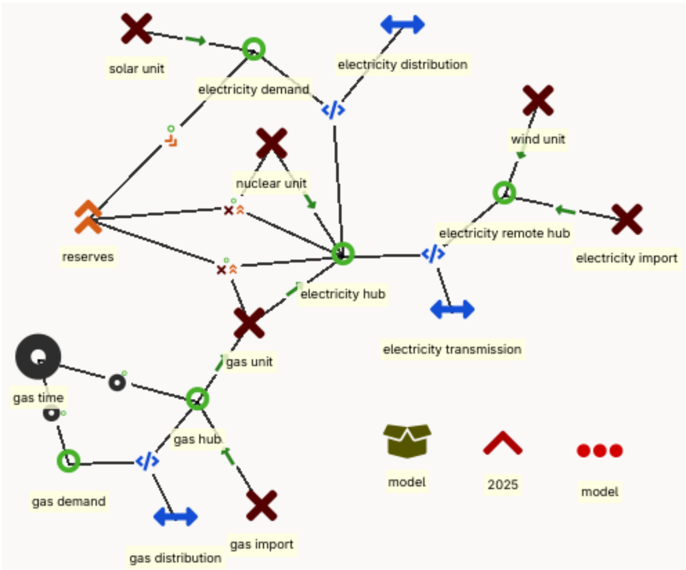

Figure 1.11: combined system

minimise:

$$
\begin{aligned}
z = &\sum_{t \in t_{gas}} \{p^{other\_operational\_costs}_{gas\_import,t} \cdot v^{flow}_{gas\_import,t} \cdot \Delta_t\} + \sum_t \{p^{other\_operational\_costs}_{electricity\_import,t} \cdot v^{flow}_{electricity\_import,t}\} \\
&+ \sum_t \{p^{startup\_cost}_{nuclear\_unit} \cdot v^{on\_up}_{nuclear\_unit,t} + p^{other\_operational\_costs}_{nuclear\_unit} \cdot v^{flow}_{nuclear\_unit,t}\} \\
&+ \sum_t \{p^{penalty}_{demand} \cdot v^{slack}_{demand,t} + p^{procurement\_cost}_{nuclear\_unit,t} \cdot v^{flow,reserve}_{nuclear\_unit,t} + p^{procurement\_cost}_{gas\_unit,t} \cdot v^{flow,reserve}_{gas\_unit,t}\} \\
&+ p^{investment\_cost}_{gas\_unit} \cdot (v^{investment}_{gas\_unit} - p^{existing\_units}_{gas\_unit}) \\
&+ p^{investment\_cost}_{wind\_unit} \cdot (v^{investment}_{wind\_unit} - p^{existing\_units}_{wind\_unit}) + p^{investment\_cost}_{solar\_unit} \cdot (v^{investment}_{solar\_unit} - p^{existing\_units}_{solar\_unit})
\end{aligned}
$$

subject to:

$$v^{state}_{storage,1} = p^{efficiency,charge}_{storage} \cdot v^{charge}_{storage,1} - v^{discharge}_{storage,1}$$

$\forall t$

$$v^{state}_{storage,t} = p^{efficiency}_{storage} \cdot v^{state}_{storage,t-1} + p^{efficiency,charge}_{storage} \cdot v^{charge}_{storage,t} - v^{discharge}_{storage,t}$$

$$p^{reserve}_{electricity\_demand,t} =< v^{flow,reserve}_{gas\_unit,t} + v^{flow,reserve}_{nuclear\_unit,t} + v^{slack}_{electricity\_demand,t}$$

$$\begin{aligned}
v^{flow,reserve}_{gas_unit,t} &\leq p^{max_reserve}_{gas_unit} \cdot v^{investment}_{gas_unit} \\
v^{flow}_{gas_unit,out,t} + v^{flow,reserve}_{gas_unit,t} &\leq p^{capacity}_{gas_unit} \cdot v^{investment}_{gas_unit} \\
v^{flow}_{nuclear_unit,t} + v^{flow,reserve}_{nuclear_unit,t} &\leq p^{capacity}_{nuclear_unit} \cdot p^{existing_units}_{nuclear_unit} \cdot v^{on}_{nuclear_unit,t}
\end{aligned}$$

$$\begin{aligned}
p^{flow_profile}_{electricity_demand,t} &= v^{flow}_{electricity_distribution,out,t} + v^{flow}_{solar_unit,t} \\
v^{flow}_{electricity_distribution,in,t} &= v^{flow}_{electricity_transport,out,t} + v^{flow}_{nuclear_unit,t} + v^{flow}_{gas_unit,out,t} \\
v^{flow}_{electricity_transport,in,t} &= v^{flow}_{wind_unit,t} + v^{flow}_{electricity_import,t}
\end{aligned}$$

$$\begin{aligned}
v^{flow}_{electricity_distribution,out,t} &= p^{efficiency}_{electricity_distribution} \cdot v^{flow}_{electricity_distribution,in,t} \\
v^{flow}_{electricity_transport,out,t} &= p^{efficiency}_{electricity_transport} \cdot v^{flow}_{electricity_transport,in,t}
\end{aligned}$$

$$\begin{aligned}
v^{flow}_{solar_unit,t} &\leq p^{flow_profile}_{solar_unit,t} \cdot p^{capacity}_{solar_unit} \cdot v^{investment}_{solar_unit} \\
v^{flow}_{wind_unit,t} &\leq p^{flow_profile}_{wind_unit,t} \cdot p^{capacity}_{wind_unit} \cdot v^{investment}_{wind_unit}
\end{aligned}$$

$$v^{flow}_{gas\_unit,out,t} \leq p^{capacity}_{gas\_unit} \cdot v^{investment}_{gas\_unit}$$

$$\begin{aligned}
v^{on}_{nuclear_unit,t} &= v^{on}_{nuclear_unit,t-1} + v^{on_up}_{nuclear_unit,t} - v^{on_down}_{nuclear_unit,t} \\
v^{on_up}_{nuclear_unit,t} &\leq 1 - v^{on}_{nuclear_unit,t} - \sum_{t'=1}^{MDT-1} v^{on_down}_{nuclear_unit,t-t'} \\
v^{on_down}_{nuclear_unit,t} &\leq v^{on}_{nuclear_unit,t} - \sum_{t'=1}^{MUT-1} v^{on_up}_{nuclear_unit,t-t'}
\end{aligned}$$

$\forall t_{gas}$

$$
\begin{aligned}
p^{flow\_profile}_{gas\_demand,t} &= v^{flow}_{gas\_distribution,out,t} \\
v^{flow}_{gas\_import,t} &= v^{flow}_{gas\_distribution,in,t} + v^{flow}_{gas\_unit,in,t} \\
v^{flow}_{gas\_distribution,out,t} &= p^{efficiency}_{gas\_distribution} \cdot v^{flow}_{gas\_distribution,in,t}
\end{aligned}
$$

$$
\begin{aligned}
v^{flow}_{gas\_unit,in,t} &\leq \frac{1}{p^{efficiency}_{gas\_unit}} \cdot p^{capacity}_{gas\_unit} \cdot v^{investment}_{gas\_unit} \\
\sum_{t \in [t_{gas}:t_{gas}+\Delta_{t_{gas}}]} v^{flow}_{gas\_unit,out,t} \cdot 1h &= p^{efficiency}_{gas\_unit} \cdot v^{flow}_{gas\_unit,in,t_{gas}} \cdot \Delta_{t_{gas}}
\end{aligned}
$$

bound to:

$$
\begin{aligned}
p^{existing\_units}_{gas\_unit} &\leq v^{investment}_{gas\_unit} && \leq p^{max\_cumulative}_{gas\_unit} \\
p^{existing\_units}_{solar\_unit} &\leq v^{investment}_{solar\_unit} && \leq p^{max\_cumulative}_{solar\_unit} \\
p^{existing\_units}_{wind\_unit} &\leq v^{investment}_{wind\_unit} && \leq p^{max\_cumulative}_{wind\_unit}
\end{aligned}
$$

$\forall t$

$$
\begin{aligned}
0 \leq & v^{slack}_{electricity\_demand,t} \\
0 \leq & v^{flow,reserve}_{gas\_unit,t} \\
0 \leq & v^{flow,reserve}_{nuclear\_unit,t} && \leq p^{max\_reserve}_{nuclear\_unit} \cdot p^{existing\_units}_{nuclear\_unit}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq & v^{flow}_{electricity\_distribution,out,t} && \leq p^{capacity}_{electricity\_distribution} \cdot p^{existing\_units}_{electricity\_distribution} \\
0 \leq & v^{flow}_{electricity\_transport,out,t} && \leq p^{capacity}_{electricity\_transport} \cdot p^{existing\_units}_{electricity\_transport}
\end{aligned}
$$

$$
0 \leq v^{flow}_{electricity\_import,t} \leq p^{capacity}_{electricity\_import} \cdot p^{existing\_units}_{electricity\_import}
$$

$$
\begin{aligned}
0 \leq & v^{flow}_{nuclear\_unit,t} && \leq p^{capacity}_{nuclear\_unit} \cdot p^{existing\_units}_{nuclear\_unit} \\
0 \leq & v^{on}_{nuclear\_unit,t} && \leq 1 \\
0 \leq & v^{on\_up}_{nuclear\_unit,t} && \leq 1 \\
0 \leq & v^{on\_down}_{nuclear\_unit,t} && \leq 1
\end{aligned}
$$

$$
0 \leq v^{flow}_{gas\_unit,out,t}
$$

$$
\begin{aligned}
0 \leq & v^{flow}_{solar\_unit,t} \\
0 \leq & v^{flow}_{wind\_unit,t}
\end{aligned}
$$

$$
\begin{aligned}
0 \leq & v^{charge}_{battery,t} && \leq p^{capacity\_charge}_{battery} \cdot p^{existing\_units}_{battery} \\
0 \leq & v^{discharge}_{battery,t} && \leq p^{capacity\_discharge}_{storage} \cdot p^{existing\_units}_{battery} \\
0 \leq & v^{state}_{battery,t} && \leq p^{capacity}_{storage} \cdot p^{existing\_units}_{battery}
\end{aligned}
$$

$\forall t_{gas}$

$$
\begin{aligned}
0 \leq & v^{flow}_{gas\_distribution,out,t} && \leq p^{capacity}_{gas\_distribution} \cdot p^{existing\_units}_{gas\_distribution} \\
0 \leq & v^{flow}_{gas\_import,t} && \leq p^{capacity}_{gas\_import} \cdot p^{existing\_units}_{gas\_import} \\
0 \leq & v^{flow}_{gas\_unit,in,t}
\end{aligned}
$$

Appendix A

Ines-spec

Below are the symbols used in the ines-spec. For a full explanation of each entity, we refer to the ines-spec repository.

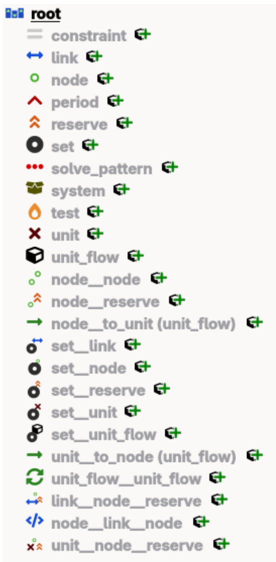

Figure A.1: Symbols used in the ines-spec.

Appendix B

Installation and use of the certification process

Installation:

1. Install Python
2. Spine Toolbox
   (a) create a python environment in a desired location, e.g. ~/spinetools/environments, assuming you opened the terminal in the spinetools folder: ‘python -m venv environments/penv‘
   (b) activate the python environment: ‘source environments/penv/bin/activate‘ (linux) ‘call environments/penv/Scripts/activate‘ (windows)
   (c) install with ‘pip install spinetoolbox‘
   (d) open spinetoolbox and set the correct environments in the settings for the tools and install plugings if necessary
3. ines tools
   (a) clone the github repository: ‘git clone https://github.com/ines-tools/ines-tools.git‘
   (b) activate the python environment (not necessary if the environment is still active)
   (c) install with ‘pip install ines-tools/.‘
4. The conversion script between ines and the energy system planning model, e.g. SpineOpt
   (a) clone the github repository: ‘git clone https://github.com/ines-tools/ines-spineopt.git‘
   (b) navigate to the folder: ‘cd ines-spineopt‘
   (c) switch to the correct branch (as the time of writing that is the new_datastructure branch): ‘git checkout new_datastructure‘

(d) go back to the spinetools folder: ‘cd ..‘

5. The ines-spec

   (a) clone the github repository: ‘git clone https://github.com/ines-tools/ines-spec.git‘

6. The energy system planning model, e.g. SpineOpt

   (a) At the time of writing, we need the developer version of SpineOpt to access a specific branch that is compatible with the conversion script. The same holds for SpineInterface. For each of these projects we need to clone the repository, navigate to the folder, select the correct branch, return to the overall folder and eventually create a julia environment

   (b) ‘git clone https://github.com/spine-tools/SpineOpt.jl.git‘

   (c) ‘cd SpineOpt.jl‘

   (d) ‘git checkout datastructure_update‘

   (e) ‘cd ..‘

   (f) ‘git clone https://github.com/spine-tools/SpineInterface.jl.git‘

   (g) ‘cd SpineInterface.jl‘

   (h) ‘git checkout #139_attempt4‘

   (i) ‘cd ..‘

   (j) install Julia with juliaup

   (k) Then we need to open the julia repl to create the environment and add the packages. The terminal is openened with the ‘julia‘ command, the package manager is accessed by typing ‘]‘ in the julia terminal.

   (l) Create and activate the environment with ‘activate environments/jenv‘

   (m) add the developer packages, starting with SpineInterface ‘dev SpineInterface.jl/.‘ and ‘dev SpineOpt.jl/.‘

   (n) Go back to the julia repl by pressing backspace and exit julia by typing ‘exit()‘

   (o) in spinetoolbox add the spineopt plugin and set the julia environment in files>settings>tools>julia to the created julia environment

Getting started:

1. Create a new project in Spine Toolbox, e.g. ’ines-certify’.

2. Add the necessary building blocks (data stores and tools) and connect them accordingly, see for example Figure 1.1.

3. For the datastore objects

   (a) create a new spine db from the data store properties window

   (b) load the template for the ines-spec and the energy system planning model

   i. for the certification setup in the ines-spec, the template is directly available in the setup itself, open the spine db editor and import the desired certification setup from the ines-spec repository (in the ines-certify folder)

   ii. for the SpineOpt model, use the load_template tool to load the template in the input spine db for the SpineOpt model

4. For the tools

   (a) add the relevant databases to the tool arguments in the tool properties window

   (b) for the conversion script open the tool editor

      i. add the main file from the ines-spineopt repository

         A. ines_to_spineopt.py

      ii. add all the other files as additional files

         A. ines_to_spineopt_entities.yaml

         B. ines_to_spineopt_entities_to_parameters.yaml

         C. ines_to_spineopt_methods.yaml

         D. ines_to_spineopt_parameters.yaml

         E. settings.yaml

5. run the project

6. Compare the results of the energy system planning model to those of the exact solution in the ines-spec repository (you can run the exact solution in the same julia environment of SpineOpt after you've explicitly added the JuMP package to that environment)

Appendix C

Extended Certification process

The certification setups from Chapter 1 are selected for a proof of concept. The full certification process consists of:

equation tests are a systematic group of tests to test each equation of energy system models individually, starting from a simple energy balance on one node with 1 demand unit and on supply unit

method tests are tests that cover multiple equations to show off a certain feature (various examples can be found in Chapter 1)

system tests are more comprehensible, similar to IEEE networks (an example can be found in the final setup of Chapter 1)

For the equation tests, inspiration can be taken from the different formulation in the proof of concept of the certification process. For the method tests, the table below is an indicative expansion of the certification process. The system tests are a point of discussion for the consortium of the ines-spec.

Table C.1: Expansion of the certification process. (*) indicate setups that are already explicitly covered in the proof of concept. (**) indicate setups that are already implicitly covered in the proof of concept.

name	description
network = copper plate	Energy balance where the demand equals the supply for a single time step. Two units where one is cheaper than the other.

network = radial	Two supply units, each connected with a directional line to the demand. The cheaper unit has the direction from the demand to the supply so the setup has no choice but to choose the more expensive unit.
(*) network = meshed	Energy balance where the demand equals the supply for a single time step. Triangular network with the two supply units and demand unit. The cheapest unit should be chosen.
(*) resolution = fixed	Same setup but with a time series for all units with the same resolution.
resolution = variable	Same setup but the time series has a variable resolution.
(*) resolution = flexible	Same setup but the two units have different resolutions (1h for expensive unit and 2h for cheap unit).
(*) stochastic	Meshed network, fixed resolution, expensive and cheap unit. The units have the same stochastics.
stochastic = flexible	Same setup but the units have different stochastics.
CO2 cap	Meshed network, fixed resolution, expensive and cheap unit. The cheap unit produces CO2 and is limited by the cap.
(*) reserves	Meshed network, fixed resolution, expensive and cheap unit. The expensive unit is able to provide reserves but not enough to entirely avoid load shedding.
cogeneration	Meshed network, fixed resolution, two supply units and two demand units but one one supply is cogeneration and one demand is for heat. The setup is heat driven as only the electricity grid has two supply units.
ramping	Meshed network, fixed resolution, expensive and cheap unit. The cheap unit is not able to ramp. The demand profile is either an elevated sine or piecewise linear _/.

flow = fixed	Meshed network, fixed resolution, expensive and cheap unit. The expensive unit is forced to produce a certain flow.
flow = limited	Meshed network, fixed resolution, expensive and cheap unit. The expensive unit is forced to produce within certain limits. This test also represents curtailment.
(**) storage = intra	Meshed network, fixed resolution, one storage unit and one supply unit. The supply unit has a fixed output that does not correspond to the demand. The storage has to be used to be feasible.
storage = inter	Same setup but there are multiple periods.
part-load efficiency	Meshed network, fixed resolution, expensive and cheap unit. The partload efficiency of the cheap unit makes it that the overall cost is higher than the expensive unit.
(**) MUT	Meshed network, fixed resolution, expensive storage and cheap unit. By setting the minimum production equal to the maximal production the output of the unit becomes fixed. By setting the minimum uptime (MUT) equal to the lifetime of the unit, whenever the unit delivers an output, that output will need to be maintained for the remainder of the simulation. That means that there will appear horizontal lines for each level of required output. To make sure that the system works, a storage unit is used to get rid of the excess power. The storage unit is very expensive such that it would be avoided if it would not be needed.
(**) MDT	Same setup, but there is a moment where the demand is zero. Normally, the plant would shut down, but due to the MDT it needs to remain on.

economic representation	Meshed network, fixed resolution, expensive and cheap unit. The numbers will be different from other setups due to the economic representation.
(*) investment = fixed	Meshed network, fixed resolution, expensive and cheap unit. Expensive unit is already invested in.
investment = cumulative	Same setup but a cumulative cap
investment = multi year	Same setup but multiple periods.
retirement = fixed	Same setup Expensive unit is already invested in but is sunset in the second period.
retirement = economic	Same setup but the retirement is not fixed.
representative periods	Meshed network, fixed resolution, expensive and cheap unit, renewable unit and intra period storage. A few representative days are chosen.
blended representative periods	Same setup but with blended representative periods and inter period storage.
(*) rolling horizon	Same setup but the periods are separated.
benders decomposition	Same setup but the investment and operation are separated from each other.
mini Europe	A miniature version of europe with very limited temporal aspects (a day = 6 hours, a season is 1 day). Each country has a demand but only its most prominent technologies. Complex setup with storage, cogeneration, CO2 cap, ramping, flow limits, part-load efficiency, MUT/MDT.